

Chapter 5 Large-scale Stochastic Impulsive Systems with Time Delay
 Details (refer to "Stability and stabilization of large-scale stochastic impulsive systems with time delay" 2019. Dynam. Continuous Discrete Impuls. Systems B Appl. Algorithm.)

Further work : Input-to-state stability for large-scale stochastic impulsive systems with state delay. Stochastic Analysis and Applications. 2023.

5.1 Problem Formulation

Interconnected system with decomposition D_i ($i=1, \dots, l$) :

$$(5.1) \begin{cases} D_i: \begin{cases} dw^i(t) = f_i(t, w_t^i) dt + g_i(t, w_t^i, w_t^e, \dots, w_t^e) dt + \sum_{j=1}^l \sigma_{ij}(t, w_t^i) dw_j(t), & t \neq \tau_k \\ \Delta w^i(\tau_k) = J_i(\tau_k, w_{\tau_k}^i), & \text{coupled} \\ w_t^i = \phi_i(s), & s \in [-\gamma, 0] \end{cases} \end{cases}$$

$t \neq \tau_k$
 $\neq j$ coupled.

where $w^i \in \mathbb{R}^{n_i}$, $f_i: \mathbb{R}_+ \times \mathbb{R}^{n_i} \rightarrow \mathbb{R}^{n_i}$, $g_i: \mathbb{R}_+ \times \mathbb{R}^n \rightarrow \mathbb{R}^{n_i}$ ($n = \sum_{i=1}^l n_i$).

$\sigma_{ij}: \mathbb{R}_+ \times \mathbb{R}^{n_j} \rightarrow \mathbb{R}^{n_i \times m_j}$. w_j — m_j -dimensional Wiener process.

$\phi_i: [-\gamma, 0] \rightarrow \mathbb{R}^{n_i}$, $J_i: T \times \mathbb{R}^{n_i} \rightarrow \mathbb{R}^{n_i}$, $T \triangleq \{\tau_k | k \in \mathbb{N}\}$ — impulsive moments.
 $0 \leq \tau_1 < \tau_2 < \dots$ and $\lim_{k \rightarrow \infty} \tau_k = \infty$. ($m = \sum_{j=1}^l m_j$)

ignoring

the interconnection

between the

subsystems

Isolated subsystem S_i :

$$(5.2) \begin{cases} S_i: \begin{cases} dw^i(t) = f_i(t, w_t^i) dt + \sigma_{ii}(t, w_t^i) dw_i(t), & t \neq \tau_k \\ \Delta w^i(\tau_k) = J_i(\tau_k, w_{\tau_k}^i) \\ w_t^i = \phi_i(s), & s \in [-\gamma, 0] \end{cases} \end{cases}$$

Goal : Find Lyapunov-like sufficient conditions under which (5.1) has uniform asymptotic stability in the m.s.

Technique : Lyapunov-Razumikhin technique. comparison method.

Results in Chapter 4.

Process : Step A: Focus on the isolated subsystem S_i , $\downarrow$ Property A
 uniform asymptotic stability of S_i Property B

Step B : Consider the interconnected part as perturbation.

Find suitable conditions to ~~make~~ guarantee uniform

stability of D_i ($i=1, \dots, l$) $\leftarrow$ Results in Chapter 4.

Notations :

$x^T = [(w^1)^T, (w^2)^T, \dots, (w^m)^T] \in \mathbb{R}^n$ — state

vector field $f: \mathbb{R}_+ \times \mathbb{R}^n \rightarrow \mathbb{R}^n$ with $f^T(t, x_t) = [f_1^T(t, w_t^1), \dots, f_m^T(t, w_t^m)]$

interconnection $g: \mathbb{R}_+ \times \mathbb{R}^n \rightarrow \mathbb{R}^n$ with $g^T(t, x_t) = [g_1^T(t, x_t), \dots, g_m^T(t, x_t)]$

diffusion matrix function $\sigma: \mathbb{R}_+ \times \mathbb{R}^n \rightarrow \mathbb{R}^{n \times m}$ with $\sigma^T(t, x_t) = [\sigma_{ij}^T(t, w_t^j)]$

Wiener process vector $W: \mathbb{R}_+ \rightarrow \mathbb{R}^m$ with $W^T = [W_1^T, \dots, W_m^T]$

impulsive functional vector $J: \mathbb{T} \times \mathbb{R}^n \rightarrow \mathbb{R}^n$ with

$$J^T(t, x_t) = [J_1^T(t, w_t^1), \dots, J_m^T(t, w_t^m)]$$

initial vector state $\Phi: [-r, 0] \rightarrow \mathbb{R}^n$ with $\Phi^T = [\phi_1^T, \dots, \phi_m^T]$

Rewrite (5.1) as follows

$$(5.3) \begin{cases} dx(t) = F(t, x_t) dt + \sigma(t, x_t) dW(t), & t \neq \tau_k \\ \Delta x(\tau_k) = J(\tau_k, x_{\tau_k}) \\ x_{t_0} = \Phi(s), & s \in [-r, 0] \end{cases}$$

where $F(t, x_t) = f(t, x_t) + g(t, x_t)$, $F \in \mathcal{L}_{ad}(\Omega; L^2[t_0, t_0+\alpha])$

$\sigma \in \mathcal{L}_{ad}(\Omega; L^2[t_0, t_0+\alpha])$, $\Phi \in L^2_{F_0}([-r, 0]; \mathbb{R}^n)$, $\alpha > 0$.

Definition of uniformly asymptotically stable in the m.s. $x \equiv 0$ of (5.3) is said to be uniformly asymptotically stable in the m.s. if

$\forall \varepsilon > 0, t_0 \in \mathbb{R}_+, \exists \delta = \delta(\varepsilon) > 0, s.t.$

$$E[\|\phi\|^2] \leq \delta \text{ implies } E[\|x(t)\|^2] < \varepsilon, \quad \forall t \geq t_0.$$

where $x(t) = x(t; t_0, \phi)$ is a solution of (5.3), $x \in PC([t_0, t_0+\alpha]; D)$ and $\phi \in L^2_{F_0}([-r, 0], D)$.

Remark :

(5.3) looks like the system in Chapter 4. But we can not use the results in Chapter 4 directly to develop Lyapunov-like conditions to ensure the stability property since the interconnection part g and $\sigma_j (i \neq j)$. That is the reason we focus on S_1 first.

Recall results in Chapter 4.

$$\begin{cases} dx(t) = f(t, x_t) dt + g(t, x_t) dW(t), & t \neq \tau_k \\ \Delta x(t) = J(t, x_{t-}) & t = \tau_k \\ x_{t_0} = \phi(s), & s \in [-r, 0] \end{cases}$$

Assumption A1. $\exists 0 \leq \rho_1 \leq \rho$ s.t. for all $\tau_k \in \mathbb{R}_+$, $x \in PC([-\tau, 0]; D)$ for some $D \subseteq \mathbb{R}^n$ (open)

if $E[\|x(\tau_k)\|^2] < \rho_1$ then $E[\|x(\tau_k)\|^2] < \rho$.

Assumption A2 For any $k \in \mathbb{N}$,

$$\tau_{\sup} = \sup\{\tau_k - \tau_{k-1}\} < \infty, \quad \tau_{\inf} = \inf\{\tau_k - \tau_{k-1}\} > 0.$$

Corollary 4.1

Assumption A1 and A2 hold. Let $V \in C^{1,2}([-\tau, \infty) \times \mathbb{R}^n; \mathbb{R}_+)$ s.t.

(i) $\exists a \in \mathcal{K}_c, b \in \mathcal{K}_v$ s.t. $\forall (t, \varphi(0)) \in [-\tau, \infty) \times S(\rho)$

$$b(\|\varphi(0)\|^2) \leq V(t, \varphi(0)) \leq a(\|\varphi(0)\|^2). \quad (\text{a.s.})$$

(ii) $\exists c \in \mathcal{K}_c, p \in PC(\mathbb{R}_+, \mathbb{R}_+)$ s.t. $\forall t \neq \tau_k$ and $\varphi \in PC([-\tau, 0]; S(\rho))$

$$\dot{V}(t, \varphi(0)) \leq -p(t) c(V(t, \varphi(0))). \quad (\text{a.s.})$$

provided that $V(t+s, \varphi(s)) \leq \bar{g}(V(t, \varphi(0)))$ for some $s \in [-\tau, 0]$ where $\bar{g} \in \mathcal{K}_3$.

(iii) $\exists \hat{g} \in \mathcal{K}_3$ s.t. at any impulsive moment $\tau_k \in T$

$$\forall \varphi \in PC([-\tau, \infty); S(\rho)).$$

$$V(\tau_k, \varphi(0)) + J(\tau_k, \varphi(\tau_k)) \leq \hat{g}(V(\tau_k, \varphi(0))). \quad (\text{a.s.})$$

with $\varphi(0) = \varphi(0)$ where $(\tau_k, \varphi(\tau_k)) \in \mathbb{R}_+ \times PC([-\tau, 0]; S(\rho))$.

(iv) $M_2 = \sup_{q \geq 0} \int_q^{\bar{g}(q)} \frac{ds}{c(s)} < \infty$ and $M_1 = \inf_{t \geq 0} \int_t^{t+\mu} p(s) ds > \mu_2$

with $\mu = \inf\{\tau_k - \tau_{k-1}\} > 0$.

Then the trivial solution $x \equiv 0$ of (4.1) is uniformly asymptotically stable in the m.s.

For S_i ($i=1, \dots, l$) Property A

$V^i \in C^{1,2}([-\tau, \infty) \times S(\rho); \mathbb{R}_+)$, $a_i, c_i \in \mathcal{K}_c$, $b_i \in \mathcal{K}_v$, $p = -\dot{V}_i > 0$ constant, $\bar{g}_i \in \mathcal{K}_3$, $\hat{g}_i \in \mathcal{K}_3$, $M_2^i < -\dot{V}_i \mu$.

← Corollary 4.1

$$f_c \triangleq \{a \in C(\mathbb{R}_+; \mathbb{R}_+) : a(0)=0, \text{ concave } \nearrow\}$$

$$f_v \triangleq \{b \in C(\mathbb{R}_+; \mathbb{R}_+) : b(0)=0, \text{ convex } \nearrow\}$$

$$k_3 \triangleq \{g \in C(\mathbb{R}_+, \mathbb{R}_+) : g(0)=0, \text{ concave } \nearrow\}$$

$$PC([a, b]; D) = \{\gamma : [a, b] \rightarrow D \mid \text{discontinuous points are at most finite, } \gamma(t^+) = \gamma(t) \text{ (right-continuous), } \gamma(t^-) \text{ exist}\}$$

Theorem 5.1 $\leftarrow$ ~~Corollary~~ Corollary 4.1

(5.3) satisfies

(i) S_i possesses Property A.

(ii) $\exists b_{ij} > 0, \bar{q} \geq 1, \text{ s.t.}$

$$g_i^T(t, \gamma) V_{\gamma^i(0)}(t, \gamma^i(0)) \leq C_i^{1/2} (\|\gamma^i(0)\|^2 \sum_{j=1}^L \bar{q} b_{ij} C_j^{1/2} (\|\gamma^j(0)\|^2))$$

(iii) $\exists e_i > 0, \text{ s.t.}$

$$(y^i)^T V_{\gamma^i(0)}(t, \gamma^i(0)) y^i \leq \bar{q} e_i \|\gamma^i(0)\|^2$$

where $y^i = \sigma_{ij}(t, \gamma^j)$

$$\rightarrow C_i (\|\gamma^i(0)\|^2)$$

(iv) $\exists d_{ij} \geq 0, \text{ s.t. } \|\sigma_{ij}(t, \gamma^j)\|^2 \leq d_{ij} C_i (\|\gamma^j(0)\|^2)$

$$\text{or } C_i = C_j \quad \forall i \neq j$$

(v) $S = [s_{ij}]_{i,j}$ is negative definite where

$$s_{ij} = \begin{cases} \alpha_i (\sigma_i + \bar{q} b_{ii}) + \frac{1}{2} \sum_{k \neq i} \bar{q} \alpha_k e_k d_{ki} & i=j \\ \frac{1}{2} \bar{q} (\alpha_i b_{ij} + \alpha_j b_{ji}) & i \neq j \end{cases}$$

for some $\alpha_i > 0$

(vi)

Then $x=0$ of (5.3) is uniformly asymptotically stable in the m.s.

Remark:

There is some mistake about the proof. But we can learn the idea and process.

Proof: Define $V(t, x) = \sum_{i=1}^L \alpha_i V^i(t, w_i)$

$$\Rightarrow (i) \sum_{i=1}^L \alpha_i b_{ii} (\|\gamma^i(0)\|^2) V(t, \gamma^i(0)) \leq \sum_{i=1}^L \alpha_i a_i (\|\gamma^i(0)\|^2)$$

For (ii) in Corollary 4.1, for $t \neq t_k$

$$\dot{V}(t, x) = \sum_{i=1}^L \alpha_i \left[\dot{V}^i(t, w_i) + g_i^T(t, x) V_{w_i}^i(t, w_i) \right]$$

$$\begin{aligned}
& + \frac{1}{2} \sum_{j=1, j \neq i}^L \text{tr} [\sigma_{ij}^T(t, w_i^i) V_{w_i^i}^i(t, w_i^i) \sigma_{ij}(t, w_i^i)] \} \\
& \leq \sum_{i=1}^L \alpha_i \left\{ \sigma_i C_i (V^i(t, w_i^i)) + C_i^{\frac{1}{2}} (\|w_i^i\|^2) \sum_{j=1}^L \bar{q}_{ij} b_{ij} C_j^{\frac{1}{2}} (\|w_j^j\|^2) \right. \\
& \quad \left. + \frac{1}{2} \sum_{j=1, j \neq i}^L \bar{q}_{ij} e_i \|\sigma_{ij}(t, w_i^i)\|^2 \right\} \\
& \leq \sum_{i=1}^L \alpha_i \left\{ \sigma_i C_i (V^i(t, w_i^i)) + C_i^{\frac{1}{2}} (\|w_i^i\|^2) \sum_{j=1}^L \bar{q}_{ij} b_{ij} C_j^{\frac{1}{2}} (\|w_j^j\|^2) \right. \\
& \quad \left. + \frac{1}{2} \sum_{j=1, j \neq i}^L \bar{q}_{ij} e_i d_{ij} C_j^{\frac{1}{2}} (\|w_j^j\|^2) \right\} \\
& \triangleq z^T S z.
\end{aligned}$$

where $z^T = (C_1^{\frac{1}{2}}(\|w^1\|^2) \dots C_L^{\frac{1}{2}}(\|w^L\|^2)) \in \mathbb{R}^L$.

S is defined in (5.3). Since S is negative-definite.

Set $\lambda_m(s)$ is the maximal eigenvalue of S , and $\lambda_m(s) < 0$.

$$\Rightarrow V(t, x) \leq \underbrace{\lambda_m(s)}_{< 0} \sum_{i=1}^L C_i (\|w_i^i\|^2) \quad C_i = C_j, i \neq j.$$

$$= \lambda_m(s) \sum_{i=1}^L C_i (\|w_i^i\|^2) \quad C \in K.$$

$$\text{For } t = \tau_k, \quad V(\tau_k, x(\tau_k)) = \sum_{i=1}^L \alpha_i V^i(\tau_k, w_i^i(\tau_k))$$

$$\text{Since } V^i(\tau_k, \varphi^i(\tau_k)) + J^i(\tau_k, \varphi^i(\tau_k)) \leq \hat{g}^i(V^i(\tau_k, \varphi^i(\tau_k)))$$

$$\Rightarrow V(\tau_k, x(\tau_k)) + J(\tau_k, x(\tau_k)) = \sum_{i=1}^L \alpha_i V^i(\tau_k, \varphi^i(\tau_k)) + J(\tau_k, \varphi^i(\tau_k))$$

$$\leq \sum_{i=1}^L \alpha_i \hat{g}^i(V^i(\tau_k, \varphi^i(\tau_k)))$$

$$\triangleq \hat{g}(V(\tau_k, x(\tau_k))).$$

Thus all the conditions of Corollary 4.1 are satisfied, therefore $x=0$ of (5.3) is uniformly asymptotically stable. $\square$

Example 5.1.

$$\begin{cases} dx = Ax dx + b f(y) dt + \sigma_{11}(x(t-1)) dw_1 + \sigma_{12}(y) dw_2, & t \neq \tau_k \\ dy = (-\xi y - \xi f(y)) dt + a^T x dt + \sigma_{21}(x) dw_1 + \sigma_{22}(y(t-1)) dw_2, & t \neq \tau_k \end{cases}$$

where $x^T = (x_1, \dots, x_4)$ — system state, $y \in \mathbb{R}$ — controller.

$A \in \mathbb{R}^{4 \times 4}$, $b \in \mathbb{R}^4$, $\xi, \tilde{\xi} \in \mathbb{R}$, $f \in \mathbb{R}$ continuous, $f(y) = 0 \Leftrightarrow y = 0$.

and $0 < y f(y) < k |y|^2$, $\sigma_{11} \in \mathbb{R}^{4 \times 1}$, $\sigma_{21} \in \mathbb{R}^{4 \times 1}$, $\sigma_{22} \in \mathbb{R}^{1 \times 1}$.

$w_1 \in \mathbb{R}^4$, $w_2 \in \mathbb{R}$.

$$\begin{cases} \Delta x(\tau_k) = f_1(\tau_k, x(\tau_k)) = \frac{1}{k^2} (-2x_1(\tau_k), -2x_2(\tau_k), 2x_3(\tau_k), 0)^T \\ \Delta y(\tau_k) = f_2(\tau_k, y(\tau_k)) = -\frac{1}{1+k^2} y(\tau_k) \end{cases}$$

$$A = \begin{pmatrix} -5 & 0 & 0 & 0 \\ 0 & -6 & 0 & 0 \\ 0 & 0 & -8 & 0 \\ 0 & 0 & 0 & -10 \end{pmatrix}, \quad \sigma_{11} = 0.01 \begin{pmatrix} \sin x_1(t-1) & 0 & \frac{x_2(t-1)}{1+x_1^2} & 0 \\ 0 & \frac{x_2(t-1)}{1+x_1^2} & 0 & -x_3^2(t-1) \\ 0 & 0 & x_3(t-1) & 0 \\ 0 & 0 & 0 & -x_4(t-1) \end{pmatrix}$$

$$b^T = (1, 1, 1, 1), \quad a^T = (1, 1, 1, 1), \quad \zeta = 5, \quad \bar{\zeta} = 2, \quad \sigma_{12} = 0.01 \frac{y}{1+y^2}, \\ \sigma_{21}^T = 0.01 (x_2, x_1, x_4, x_3), \quad \sigma_{22} = 0.01 \sin y(t-1)$$

$$\text{Let } V^1(x) = \|x\|^2, \quad V^2(y) = y^2$$

$$\begin{cases} dx = Ax dt + \sigma_{11}(x(t-1)) dw_1 \\ dy = (-\zeta y - \zeta f(y)) dt + \sigma_{22}(y(t-1)) dw_2 \end{cases} \quad \begin{matrix} S_1 \\ S_2 \end{matrix}$$

$$\begin{aligned} L_1 V^1(x) &= 2x^T A x + \frac{1}{2} \text{tr} [\sigma_{11}^T(x(t-1)) \cdot 2 \cdot \sigma_{11}(x(t-1))] \\ &\leq -10 \|x\|^2 + \frac{1}{2} \text{tr} \left[\sigma_{11}^T(x(t-1)) \cdot 2 \cdot \sigma_{11}(x(t-1)) \right] \\ &\leq (-10 + 0.0002 \bar{q}) \|x\|^2 \end{aligned}$$

$$\begin{aligned} L_2 V^2(y) &= -2\zeta y^2 - 2\zeta y f(y) + \frac{1}{2} \cdot 2 \cdot \sigma_{22}^2(y(t-1)) \\ &= -2\zeta y^2 - 2\zeta y f(y) + 0.00001 \sin^2 y(t-1) \\ &\leq (-2\zeta + 0.0000 \bar{q}) y^2 = (-10 + 0.0000 \bar{q}) y^2 \end{aligned}$$

$$\sigma_1 = -10 + 0.0002 \bar{q}, \quad \sigma_2 = -10 + 0.0000 \bar{q}, \quad \bar{q} = 2$$

$$\begin{aligned} V_{x^T}^1(x) g_1(x, y) &= 2x^T b f(y) \leq 4 \|x\| \|f(y)\| \\ &\leq 4 \|x\| k |y| \\ V_{y^T}^2(y) g_2(x, y) &= 2y \cdot a^T x \leq 4 \|x\| |y| \end{aligned}$$

$$\frac{1}{2} \text{tr} (\sigma_{12}^T V_{xx}^1 \sigma_{12}) = 0.00001 y^2, \quad \frac{1}{2} \text{tr} (\sigma_{21}^T b V_{yy}^2 \sigma_{21}) = 0.00001 \|x\|^2$$

$$\begin{aligned}
 \underline{1} \quad V(x, y) &= \alpha_1 [L_1 V_1(x) + g_1^T(x) V_{x_1}^T(x)] + \alpha_2 [L_2 V_2(y) + g_2^T(y) V_{y_1}^T(y)] \\
 &\quad + \alpha_1 \frac{1}{2} \text{tr}[\Sigma_1 \Sigma_1^T V_{x_1 x_1}^T(x) \Sigma_1^T(y)] + \alpha_2 \frac{1}{2} \text{tr}[\Sigma_2 \Sigma_2^T(x) V_{y_1 y_1}^T(y) \Sigma_2^T(y)] \\
 &\leq \alpha_1 \Sigma_1 \|x\|^2 + \alpha_1 4k \|x\| \|y\| + \alpha_2 \Sigma_2 \|y\|^2 + \alpha_2 4 \|x\| \|y\| \\
 &\quad + \alpha_1 0.00001 y^2 + \alpha_2 0.0000 \|x\|^2 \\
 &= (x^T, y) \cdot \begin{pmatrix} \alpha_1 \Sigma_1 + \alpha_2 0.00001 & 2\alpha_1 k + 2\alpha_2 \\ 2\alpha_1 k + 2\alpha_2 & \alpha_2 \Sigma_2 + \alpha_1 0.00001 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} \quad x: a_{ij} y_j \\
 &= S
 \end{aligned}$$

Fix $\alpha_1 = 1, \alpha_2 = 1$

$$S = \begin{pmatrix} -9.9995 & 2k+2 \\ 2k+2 & -9.9997 \end{pmatrix} \quad \text{negative definite for } k < 3.9998$$

$\lambda_1(s) = -15.082$ $\lambda_2(s) = -2.917$ $k=2$

$$\begin{aligned}
 V(x(\tau_k), y(\tau_k)) &= \|x(\tau_k)\|^2 + y^2(\tau_k) \\
 &\leq \left(1 + \frac{2}{k^2}\right) \|x(\tau_k)\|^2 + \left(1 - \frac{5}{1+k^2}\right) y^2(\tau_k) \\
 &\leq \left(1 + \frac{2}{k^2}\right) (\|x(\tau_k)\|^2 + y^2(\tau_k)) \\
 &= \left(1 + \frac{2}{k^2}\right) V(x(\tau_k), y(\tau_k))
 \end{aligned}$$

Choosing $\mu = 0.04, k=2$. For isolated system
 $a_i(s) = b_i(s) = s^2 \rightarrow$ ensure the asymptotic stability

in the m.s. of isolated subsystems.

The eigenvalue of matrix S are $-15.082, -2.917$.

Choose $\delta = 2.917$ and $\mu > 0.14 \Rightarrow (x, y)^T = (0, 0)$ is exponentially stable in the m.s.